

25. Where to Shop?, continued Refer to Exercise 24.

The printout that follows provides the average costs of the selected items for the $k = 4$ stores.

Means

Store	N	Price
1. Ralphs	8	4.34000
2. Stater Bros	8	3.85250
3. Vons	8	4.13500
4. WinCo	8	2.68125

- What is the appropriate value of $q_{\alpha}(k, df)$ for testing for differences among stores?
- What is the value of $\omega = q_{\alpha}(k, df) \sqrt{\frac{MSE}{b}}$?
- Use Tukey's pairwise comparison test among stores used to determine which stores differ significantly in average prices of the selected items.

a. $q_{0.05}(4, 24) = 3.942$

b. $\omega = 3.942 \cdot \sqrt{\frac{MSE}{b}} = 3.942 \cdot \sqrt{\frac{0.1658}{8}} = 0.5675$

c. $|\bar{X}_R - \bar{X}_S| = 0.4875 < 0.5675 \Rightarrow$ aren't significant different

$|\bar{X}_R - \bar{X}_V| = 0.205 < 0.5675 \Rightarrow$

$|\bar{X}_S - \bar{X}_V| = 0.2825 < 0.5675 \Rightarrow$ aren't

$|\bar{X}_R - \bar{X}_W| = 1.65875 > 0.5675 \Rightarrow$ are significant different

$|\bar{X}_S - \bar{X}_W| = 1.17125 > 0.5675 \Rightarrow$ are

$|\bar{X}_V - \bar{X}_W| = 1.45375 > 0.5675 \Rightarrow$ are

W is significantly different from S, V, R in price average of the select items